

Reliability-Aware Adaptive Contention Control for EMLSR Multi-Link Operation in IEEE 802.11bn WLANs

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Abstract—IEEE 802.11bn Ultra High Reliability (UHR) extends multi-link operation (MLO) toward a reliability-oriented medium-access control (MAC), but multi-link contention under the Enhanced Multi-Link Single-Radio (EMLSR) active-link constraint resists closed-form analysis. We study three-band multi-link devices (MLDs) that coexist with single-link legacy stations, and develop a Markov/fixed-point closure for their MAC-layer behavior. The closure is exact under saturated, per-link-independent simultaneous transmit and receive (STR); it reduces to an availability-based approximation under saturated EMLSR, with a validity range made explicit in the transition and synchronization costs; and it extends to unsaturated traffic through per-link Bernoulli arrivals. On top of this closure we build a reliability-aware controller that adapts the per-link minimum contention window from MAC observables—collision rate, blocked ratio, queue backlog, 95th-percentile head-of-line (HOL) excess, and delivery deficit—within the standard EDCA/EMLSR loop, so its operating regime is analytically auditable rather than empirical. Against single-link-device (SLD), STR-MLD, and static-EMLSR baselines, the controller raises throughput by 10.6%, lowers 95th-percentile HOL-service delay by 28%, and improves Jain’s fairness from 0.910 to 0.954 at 48 stations, recovering most of the SLD reliability ceiling without sacrificing MLD link aggregation. A triangulation across our analytical model, a discrete-event simulator, and ns-3 closes within a 2–4% mean relative gap over MCS, legacy-ratio, payload, multi-band, and offered-load sweeps.

Index Terms—IEEE 802.11bn, ultra high reliability (UHR), Wi-Fi 8, MLO, EMLSR, medium access control (MAC), performance modeling, Markov chain, reliability-aware contention control

I. Introduction

IEEE 802.11bn Ultra High Reliability (UHR; informally Wi-Fi 8) targets higher throughput under stress, lower 95th-percentile latency, and reduced MPDU loss for isolated and overlapping BSSs [1], [2], extending 802.11be multi-link operation (MLO) toward a reliability-oriented medium-access control (MAC) [3]–[7].

Multi-link contention is markedly harder to analyze than single-link DCF/EDCA. Each link is an independent contention domain, yet at the multi-link device (MLD)

level one must also account for queue coupling, link selection, the Enhanced Multi-Link Single-Radio (EMLSR) active-link constraint, and transition and synchronization costs [7]–[17]. Single-link Bianchi-family analyses [18]–[22], [24]–[30] therefore do not transfer directly, and recent MLO studies [7]–[17], [23] rarely separate, within one framework, the regimes that admit an exact statement from those that must remain approximate.

We close this gap for three-link MLDs coexisting with legacy STAs. We pair a reliability-aware adaptive contention controller, which runs inside the standard EDCA/EMLSR loop, with a Markov/fixed-point closure that delivers exact STR, an availability-based EMLSR approximation with an explicit validity range, and a per-link Bernoulli unsaturated extension whose outputs the controller consumes as observables. Our contributions are as follows.

- A reliability-aware adaptive-CW controller that adds no new synchronization or segmentation primitives, yet at 48 STAs raises throughput by 10.6%, lowers p95 HOL-service delay by 28%, and lifts Jain’s fairness from 0.910 to 0.954 over static EMLSR.
- An availability-coupled EMLSR fixed point that routes the single-active-radio constraint through both a bidirectional attempt product and an availability-scaled effective contender count.
- A Bonferroni first-order regime bound (Lemma 1) that audits the controller’s operating envelope.
- A triangulation of analytical model, discrete-event simulator, and ns-3 that closes within a 2–4% mean relative gap.
- A dominance audit against SLD, STR-MLD, and static-EMLSR baselines across MCS, legacy-ratio, payload, and asymmetric multi-band sweeps, separating EMLSR’s reliability- and throughput-oriented regimes.

Throughout, the contribution is positioned at the MAC layer: physical-layer effects enter only through fixed per-link MCS/rate and timing constants, while contention, access delay, and reliability are the objects of modeling and control.

II. Related Work

Single-link DCF/EDCA analysis rests on Bianchi’s saturated Markov model [18], [25], with the 802.11 contention

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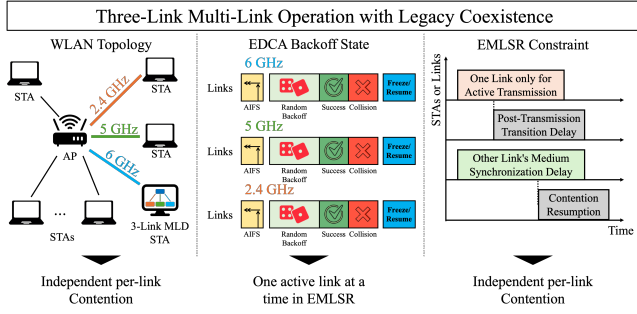


Fig. 1. Target system model with three-link MLDs coexisting with per-link legacy STAs.

rules [24], the per-cycle throughput limit [26], and extensions for EDCA priority/delay [19], unsaturated throughput [20], [27], queueing/access-delay [28], refined modelling [21], and unified stability/delay [22], a single-link MAC performance-modeling lineage we extend to multi-link EMLSR contention; none address three-link MLDs coexisting with legacy STAs under an EMLSR active-link constraint. The 802.11ax baseline [29], [30] carries OFDMA/MU-MIMO but not multi-link contention. MLO surveys position it as a core Wi-Fi 7/802.11be feature [3]–[7], a broad literature studies MLO traffic distribution, throughput, latency, and coexistence [7]–[12], [23], and EMLSR-focused works expose switching/synchronisation overhead, backoff overflow, and fairness [13]–[17]. Closest to ours, [15] compares EMLSR/SLD/STR MLD at the device level, [11] analyses MLO delay under finite load, and [16], [17] propose synchronous/data-division enhancements; none expose the exact-vs-approximate boundary in one framework or validate against ns-3. We unify exact STR, a validity-range EMLSR approximation (Lemma 1), a Bernoulli unsaturated extension, and a UHR-aligned adaptive contention controller in one Markov/fixed-point closure.

III. System Model

We consider a WLAN environment in which three-link multi-link device (MLD) stations capable of using the 2.4 GHz, 5 GHz, and 6 GHz bands coexist with single-link legacy stations (STAs). Each link forms an independent contention domain, and channel access is carried out via per-link EDCA backoff contention. A legacy station contends on only one link, whereas an MLD station acts as a multi-link contender with access to all three links. This corresponds to a single mixed-BSS analytical core used to explain UHR-oriented MAC reliability, rather than directly addressing the wider isolated/overlapping BSS scope covered by the P802.11bn UHR public documents [1], [2].

Figure 1 summarises the contention topology: a three-link MLD joins all three 2.4/5/6 GHz domains while a legacy STA is single-link, combining independent per-link contention with inter-link coupling at the MLD level. We treat as standard-based the per-link CW/AIFS/backoff

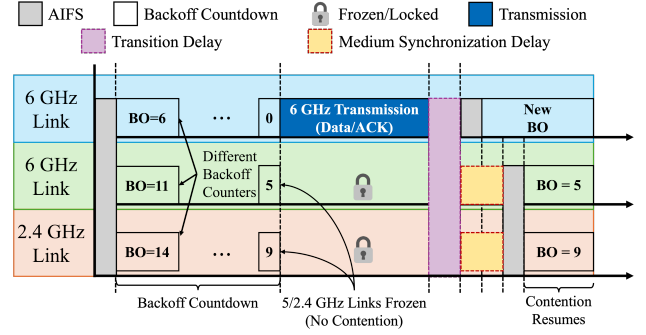


Fig. 2. Time alignment of EMLSR active link, transition, and synchronization.

and success/collision/freeze rules, the EMLSR active-link mode (exchange around one active link with T_{tr}/T_{sync} before and after), and the legacy-vs-MLD contender distinction. Under STR, per-link backoff and TXOPs proceed independently and several links may start together; under EMLSR only one active link carries the exchange while the others wait on an availability constraint—so STR is the exact baseline and EMLSR the availability-based approximation target.

Figure 2 shows, after a link is chosen as the active link in EMLSR, the order in which the other links go through block, transition, and synchronization recovery before returning to contention. Because of this structure, EMLSR does not close under the direct product of per-link Bianchi chains, and a fixed-point closure with availability and blocking fraction is required.

Two elements are research-oriented abstractions: the shared HOL queue coupling (not an IEEE 802.11 standard feature; an implementation-level queue assumption) and a slot-time service model that does not fully replicate detailed PHY control-frame timing. The paper therefore builds a tractable framework on top of the standard contention structure rather than replicating the standard implementation. Saturated per-link-independent STR is the exact-analytic reference, saturated EMLSR is the approximate-analytic reference, and the unsaturated regime is the per-link Bernoulli extension; exact total queue delay under shared HOL, and detailed external-simulator MAC-queue semantics, are out of scope.

Proposition 1 (Conditions for STR direct-product exactness). The direct-product exactness holds iff (i) each link has a saturated, independent HOL packet, (ii) there is no shared HOL packet or cross-link scheduler, (iii) there are no active-link constraints and no transition/synchronization timers, and (iv) backoff freeze/resume is determined purely by the contention domain of the same link; hence the shared HOL abstraction or an EMLSR access constraint invalidates the direct-product as an exact claim.

IV. Analytical Model

The analytical model is organised as a four-layer pipeline (a one-page summary diagram is provided in

TABLE I
Key notation for the analytical model.

Symbol	Meaning
$\mathcal{L}=\{2,4,5,6\}$	three links (bands)
$n_\ell^{(L)}, n_\ell^{(M)}$	legacy STAs on ℓ ; number of MLDs
$\tau_\ell^{(L)}, \tau_\ell^{(M)}$	per-class attempt probability on ℓ
$p_\ell^{(L)}, p_\ell^{(M)}$	per-class conditional collision probability
α_ℓ, β_ℓ	MLD availability and blocking fraction on ℓ
$\nu_\ell = \alpha_\ell n_\ell^{(M)}$	effective MLD contender count on ℓ
g_j	per-basic-slot MLD transmission-start rate on j
$H_{j \rightarrow \ell}$	mean interval by which link j blocks ℓ
T_{tr}, T_{sync}	EMLSR transition and synchronization cost
μ_ℓ	HOL service rate; $\mathbb{E}[D_{HOL}] = \sigma/\mu_\ell$

the supplementary material). Layer 1 is the single-link exact per-link chain (countdown, success-reset, collision-increment). Layer 2 composes three such chains into the direct product that is exact under independent per-link STR saturation (Proposition 1). Layer 3 is the availability-based EMLSR fixed point: tentative attempts $\{\tau_\ell^{(L)}, \tau_\ell^{(M)}\}$ feed the effective contender count $\nu_\ell = \alpha_\ell n_\ell^{(M)}$ and the bidirectional-product attempt $\lambda_\ell^{(M)}$, which drive the slot events $\{P_{idle}, P_s^{(L)}, P_s^{(M)}, P_c\}$ and a mean slot $\mathbb{E}[T_{slot,\ell}]$ that charges the $T_{tr} + T_{sync}$ overhead on MLD successes, while β_ℓ and $\alpha_\ell = 1 - \beta_\ell$ close the coupled fixed point. Layer 4 is the unsaturated Bernoulli extension that reads $P_s^{(c,b)} = P_s^{(c)}/\mathbb{E}[T_{slot}]$ from Layer 3 to form the service rate $\mu_\ell^{(c)}$, utilization $\rho_\ell^{(c)}$, HOL-service delay $\sigma/\mu_\ell^{(c)}$, and queue delay. The exact STR, approximate EMLSR, and unsaturated regions share one skeleton.

Notation. Let $\mathcal{L} = \{2,4,5,6\}$ (1), with $n_\ell^{(L)}$ legacy STAs per link, $n_\ell^{(M)}$ MLDs, and total contenders $n_\ell = n_\ell^{(L)} + n_\ell^{(M)}$ (2). The MAC parameters are CW_{min} , CW_{max} , defer A , success/collision lengths T_s/T_c , basic-slot duration σ , and payload L ; all of $A, T_s, T_c, T_{tr}, T_{sync}$ are in basic slots and time-unit delay is obtained by multiplying by σ . The busy ratio accounts only for success+collision occupancy; the post-busy AIFS is not included.

$$\mathcal{L} = \{2,4,5,6\} \quad (1)$$

$$n_\ell = n_\ell^{(L)} + n_\ell^{(M)} \quad (2)$$

A. Exact STR Model under Saturated Per-Link Independence

Under saturated per-link-independent STR, the state of a tagged contender on link ℓ is expressed as a two-dimensional state consisting of the retry stage and the backoff counter,

$$X_\ell(t) = (s_\ell(t), b_\ell(t)) \quad (3)$$

The per-stage contention window is defined as

$$W_i = CW_i + 1 \quad (4)$$

Then, in steady state, the attempt probability τ_ℓ and conditional collision probability p_ℓ on link ℓ are determined as the fixed point of

$$\tau_\ell = \frac{2}{(1 - p_\ell) \sum_{i=0}^{m-1} p_\ell^i (W_i + 1) + p_\ell^m (W_m + 1)} \quad (5)$$

¹ and

$$p_\ell = 1 - (1 - \tau_\ell)^{n_\ell - 1} \quad (6)$$

That is, the core of the exact STR model is to apply the single-link Bianchi structure given by (5) and (6) per link, and then combine the contention processes of the three links into a direct-product form.

The generic-slot probabilities are $P_{idle,\ell} = (1 - \tau_\ell)^{n_\ell}$ (7), $P_{s,\ell} = n_\ell \tau_\ell (1 - \tau_\ell)^{n_\ell - 1}$ (8), $P_{c,\ell} = 1 - P_{idle,\ell} - P_{s,\ell}$ (9), mean slot $\mathbb{E}[T_{slot,\ell}] = P_{idle,\ell} + P_{s,\ell}(T_s + A) + P_{c,\ell}(T_c + A)$ (10), and basic-slot $\tau_\ell^{(b)} = \tau_\ell / \mathbb{E}[T_{slot,\ell}]$, $P_{s,\ell}^{(b)}$, $P_{c,\ell}^{(b)}$ defined analogously (11).

$$P_{idle,\ell} = (1 - \tau_\ell)^{n_\ell} \quad (7)$$

$$P_{s,\ell} = n_\ell \tau_\ell (1 - \tau_\ell)^{n_\ell - 1} \quad (8)$$

$$P_{c,\ell} = 1 - P_{idle,\ell} - P_{s,\ell} \quad (9)$$

$$\mathbb{E}[T_{slot,\ell}] = P_{idle,\ell} + P_{s,\ell}(T_s + A) + P_{c,\ell}(T_c + A) \quad (10)$$

$$P_{s,\ell}^{(b)} = P_{s,\ell} / \mathbb{E}[T_{slot,\ell}], \quad P_{c,\ell}^{(b)} = P_{c,\ell} / \mathbb{E}[T_{slot,\ell}] \quad (11)$$

With $\mathbf{X}(t) = (X_{2,4}(t), X_5(t), X_6(t))$ and local transition kernels K_ℓ , the global transition factorises as $P_{\mathbf{x} \rightarrow \mathbf{x}'} = \prod_\ell K_\ell(x_\ell, x'_\ell)$ (12) under the four conditions of Proposition 1; a shared HOL packet or inter-link scheduler couples backoff start times across links and breaks this factorisation.

$$P_{\mathbf{x} \rightarrow \mathbf{x}'} = \prod_{\ell \in \mathcal{L}} K_\ell(x_\ell, x'_\ell) \quad (12)$$

Per-link metrics are throughput $S_\ell = P_{s,\ell} L / (\mathbb{E}[T_{slot,\ell}] \sigma)$ ($S_{total} = \sum_\ell S_\ell$) (13), busy ratio $\rho_\ell^{(busy)} = (P_{s,\ell} T_s + P_{c,\ell} T_c) / \mathbb{E}[T_{slot,\ell}]$ (14), and HOL-service rate $\mu_\ell = P_{s,\ell}^{(b)} / n_\ell$ with $\mathbb{E}[D_{\ell,HOL}] = \sigma / \mu_\ell$ (15).

$$S_\ell = \frac{P_{s,\ell} L}{\mathbb{E}[T_{slot,\ell}] \sigma}, \quad S_{total} = \sum_\ell S_\ell \quad (13)$$

$$\rho_\ell^{(busy)} = \frac{P_{s,\ell} T_s + P_{c,\ell} T_c}{\mathbb{E}[T_{slot,\ell}]} \quad (14)$$

$$\mu_\ell = \frac{P_{s,\ell}^{(b)}}{n_\ell}, \quad \mathbb{E}[D_{\ell,HOL}] = \frac{\sigma}{\mu_\ell} \quad (15)$$

¹The (W_i+1) multiplier reflects a slot-counting convention that bookkeeps the post-success deferral slot jointly with the backoff window, consistent across all stages and the same convention used by the internal simulator's contender update; restoring the original Bianchi W_i multiplier shifts τ by 1–5 % at small n ($n \leq 3$) and below 1 % at $n \geq 10$, propagating to a <1 % throughput offset and leaving the validity-range and cross-method conclusions in §V–VI unchanged.

Under STR, the MLD effective service rate aggregates as $\mu_{\text{MLD,eff}}^{\text{STR}} = \sum_{\ell} \mu_{\ell}^{(M)}$ (16); under EMLSR, simultaneous service is impossible, so per-link $\mu_{\ell}^{(M)}$ is not a node-wide rate but a per-link HOL service-interval metric.

$$\mu_{\text{MLD,eff}}^{\text{STR}} = \sum_{\ell \in \mathcal{L}} \mu_{\ell}^{(M)} \quad (16)$$

B. Approximate EMLSR Model under Saturation

Under EMLSR, a successful exchange on one link shortens the accessible time on the other links, so the exact direct-product structure that held under STR no longer persists. Writing it exactly would require a high-dimensional state vector that includes the active link, the release timer, and per-link synchronization recovery timers. This paper absorbs that complexity into an availability-based approximation.

We first separate the legacy class and the MLD class and define

$$\tau_{\ell}^{(L)} = f(p_{\ell}^{(L)}), \quad \tau_{\ell}^{(M)} = f(p_{\ell}^{(M)}) \quad (17)$$

Here $f(\cdot)$ follows the finite-stage Bianchi form of (5), and in our default setting the legacy and MLD classes share the same $\text{CW}_{\min}$, $\text{CW}_{\max}$, and AIFS. We then fix a priority function $\pi(\ell)$ with $\pi(6) = 1$, $\pi(5) = 2$, $\pi(2,4) = 3$, and let the higher-priority link set be

$$\mathcal{H}(\ell) = \{j \in \mathcal{L} : \pi(j) < \pi(\ell)\} \quad (18)$$

This priority does not represent a physical preference among bands; it is an analytically fixed, deterministic tie-breaking order used when the same MLD reaches zero backoff on several eligible links simultaneously, so (18) is not a standard-mandated PHY priority but an analytical closure that reflects the simulator's default link-selection order. Letting α_{ℓ} denote the MLD availability on link ℓ , and introducing the link-selection factor letting $\alpha_{\ell} \in [0, 1]$ denote the fraction of time an MLD is in backoff on link ℓ , the MLD tentative attempt rate is approximated under the single-active-radio constraint by the bidirectional link-competition product

$$\lambda_{\ell}^{(M)} = \alpha_{\ell} \tau_{\ell}^{(M)} \prod_{j \in \mathcal{L} \setminus \{\ell\}} (1 - \alpha_j \tau_j^{(M)}) \quad (19)$$

The bidirectional product $\prod_{j \neq \ell} (1 - \alpha_j \tau_j^{(M)})$ captures, to first order, the fact that only one of the MLD's $|\mathcal{L}|$ per-link backoff counters wins the race in any slot. No separate link-selection factor ϕ_{ℓ} is needed: α_{ℓ} itself smoothly interpolates between the sparse regime ($\alpha \approx 1$) and the dense regime ($\alpha \rightarrow 0$).

Once an MLD wins on link $j \neq \ell$ for the next TXOP it is no longer a contender on ℓ , so the slot-level MLD contender count on ℓ is not the full population $n^{(M)}$ but the effective count

$$\nu_{\ell} = \alpha_{\ell} n^{(M)}. \quad (20)$$

The class-specific self-consistency collision probabilities then close as

$$p_{\ell}^{(L)} = 1 - (1 - \tau_{\ell}^{(L)})^{\nu_{\ell}^{(L)} - 1} \cdot (1 - \tau_{\ell}^{(M)})^{\nu_{\ell}} \quad (21)$$

$$p_{\ell}^{(M)} = 1 - (1 - \tau_{\ell}^{(L)})^{\nu_{\ell}^{(L)}} \cdot (1 - \tau_{\ell}^{(M)})^{\nu_{\ell} - 1} \quad (22)$$

so the single-active-radio constraint enters both through the bidirectional product in (19) and through the effective contender count $\nu_{\ell} = \alpha_{\ell} n^{(M)}$.

Availability is expressed via the blocking fraction β_{ℓ} as

$$\alpha_{\ell} = 1 - \beta_{\ell} \quad (23)$$

The mean exchange duration of source link j is

$$\bar{D}_j = (1 - p_j^{(M)}) T_{s,j} + p_j^{(M)} T_{c,j} \quad (24)$$

and the blocking fraction sums only over source links other than ℓ (an MLD transmitting on ℓ itself does not “block” ℓ from its own backoff resumption—that cost is already folded into the mean slot of ℓ , (34)):

$$\beta_{\ell} \approx \sum_{j \in \mathcal{L} \setminus \{\ell\}} g_j H_{j \rightarrow \ell} \quad (25)$$

Here g_j is the per-basic-slot effective MLD transmission-start rate on source link j , not the success rate; collision-induced blocking is absorbed into $\bar{D}_j$ in (24). The average length by which source link $j \neq \ell$ blocks link ℓ is

$$H_{j \rightarrow \ell} = \bar{D}_j + T_{\text{tr},j} + T_{\text{sync},j} \quad (26)$$

Here $T_{\text{tr},j}$ is a source-link-specific transition cost, and $T_{\text{sync},j}$ is the medium synchronization cost imposed on the other links after transmission by source link j . That is, we model synchronization cost not as a victim-link-specific recovery constant but as a source-link-specific post-exchange overhead that a source-link event induces on the remaining links. Therefore, (23), (24), (25), and (26) constitute the core of the availability closure.

More precisely, the exact blocking fraction must be the probability of the “union of events that block link ℓ ,” which has an inclusion–exclusion structure. We keep only the first-order term:

$$\begin{aligned} \beta_{\ell}^{\text{exact}} &= \Pr\left(\bigcup_{j \in \mathcal{L}} B_{j \rightarrow \ell}\right) \\ &= \sum_j \Pr(B_{j \rightarrow \ell}) - \sum_{j < k} \Pr(B_{j \rightarrow \ell} \cap B_{k \rightarrow \ell}) + \dots \end{aligned} \quad (27)$$

is approximated by

$$\beta_{\ell}^{\text{1st}} = \sum_{j \in \mathcal{L}} g_j H_{j \rightarrow \ell} \quad (28)$$

The remainder, ignoring overlap terms, is approximately

$$\beta_{\ell}^{\text{exact}} - \beta_{\ell}^{\text{1st}} = \mathcal{O}\left(\left(\sum_{j \in \mathcal{L}} g_j H_{j \rightarrow \ell}\right)^2\right) \quad (29)$$

so the approximation is most justified where (37) stays small—a first-order closure with overlap dropped, not a tight bound.

The class-wise slot events on link ℓ , with $\nu_{\ell} = \alpha_{\ell} n^{(M)}$ from (20), follow eqs. (30)–(33), and the mean slot length $\mathbb{E}[T_{\text{slot},\ell}]$ (34) extends the STR form by charging the single-radio overhead $T_{\text{tr},\ell} + T_{\text{sync},\ell}$ only on MLD success

events. This is the source-link dual of (26): on the victim-link side the same interval is absorbed by $H_{j \rightarrow \ell}$. Aggregate throughput then reduces to a link-wise sum (35), since α_ℓ has internalised the single-radio constraint via the effective contender count:

$$P_{\text{idle},\ell} = (1 - \tau_\ell^{(L)})n_\ell^{(L)}(1 - \tau_\ell^{(M)})^{\nu_\ell} \quad (30)$$

$$P_{s,\ell}^{(L)} = n_\ell^{(L)}\tau_\ell^{(L)}(1 - \tau_\ell^{(L)})n_\ell^{(L)-1}(1 - \tau_\ell^{(M)})^{\nu_\ell} \quad (31)$$

$$P_{s,\ell}^{(M)} = \nu_\ell\tau_\ell^{(M)}(1 - \tau_\ell^{(M)})^{\nu_\ell-1}(1 - \tau_\ell^{(L)})n_\ell^{(L)} \quad (32)$$

$$P_{c,\ell} = 1 - P_{\text{idle},\ell} - P_{s,\ell}^{(L)} - P_{s,\ell}^{(M)} \quad (33)$$

$$\begin{aligned} \mathbb{E}[T_{\text{slot},\ell}] &= P_{\text{idle},\ell} + P_{s,\ell}^{(L)}(T_{s,\ell} + A) + P_{c,\ell}(T_{c,\ell} + A) \\ &\quad + P_{s,\ell}^{(M)}(T_{s,\ell} + A + T_{\text{tr},\ell} + T_{\text{sync},\ell}) \end{aligned} \quad (34)$$

$$S_{\text{total}} = \sum_\ell \frac{(P_{s,\ell}^{(L)} + P_{s,\ell}^{(M)})L}{\mathbb{E}[T_{\text{slot},\ell}]\sigma} \quad (35)$$

The single-radio constraint enters only through (19) and (20).

We solve (17)–(33) together with (23) by damped fixed-point iteration: initialise $\alpha_\ell^{(0)} = 1$, $g_\ell^{(0)} = 0$ and at each step compute provisional $\tilde{\lambda}_\ell$, $\tilde{p}_\ell^{(c)}$, $\tilde{\tau}_\ell^{(c)}$, slot events $\{P_{\text{idle}}, P_s^{(L)}, P_s^{(M)}, P_c\}$, basic-slot $\tilde{g}_\ell$, and $\tilde{\beta}_\ell, \tilde{\alpha}_\ell$ from (19)–(33) and (24)–(26), then apply the damped update

$$\begin{aligned} x^{(k+1)} &\leftarrow (1 - \eta)x^{(k)} + \eta\tilde{x}^{(k+1)}, \quad \eta = 0.2, \\ x &\in \{\tau_\ell^{(L)}, \tau_\ell^{(M)}, \alpha_\ell\}, \end{aligned} \quad (36)$$

iterating until $\max_\ell |\alpha_\ell^{(k)} - \alpha_\ell^{(k-1)}| < \varepsilon$ or the cap $K_{\text{max}} = 2 \times 10^4$ is reached. We refer to this procedure as Algorithm 1 below for cross-reference.

Because (23) is a first-order closure that ignores overlap among blocking events, its most natural applicability condition is the moderate-density operating region,

$$\sum_{j \in \mathcal{L}} g_j H_{j \rightarrow \ell} \ll 1 \quad (37)$$

We therefore interpret the EMLSR approximation not as an exact predictor but as a validity-range approximation that holds in regions where (37) is not excessively violated. Lemma 1 (First-order regime bound). Let $c_\ell \triangleq \sum_{j \in \mathcal{L}} g_j H_{j \rightarrow \ell}$ and assume $c_\ell < 1$ for every $\ell \in \mathcal{L}$. Then the absolute error between the first-order blocking fraction $\beta_\ell^{1\text{st}}$ of (28) and the exact union-of-blocking-events probability $\beta_\ell^{\text{exact}}$ of (27) is bounded by

$$0 \leq \beta_\ell^{1\text{st}} - \beta_\ell^{\text{exact}} \leq \frac{1}{2}c_\ell^2. \quad (38)$$

Sketch. By Bonferroni's inequality applied to the union $\bigcup_j B_{j \rightarrow \ell}$ with per-source probabilities $\Pr(B_{j \rightarrow \ell}) \approx g_j H_{j \rightarrow \ell}$, the exact union is bounded by the first moment minus at most $\sum_{j < k} \Pr(B_{j \rightarrow \ell}) \Pr(B_{k \rightarrow \ell}) \leq \frac{1}{2}c_\ell^2$, so the first-order closure overestimates β_ℓ by at most $c_\ell^2/2$; this is the structural reason why our EMLSR approximation remains quantitatively usable in the regimes reported in Section VI. $\square$

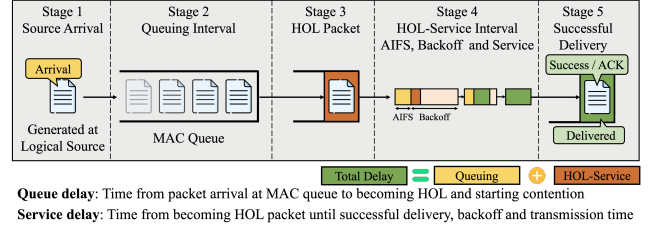


Fig. 3. Queueing interval, HOL-service interval, and total delay used in the simulator and the unsaturated approximation.

We do not claim a global contraction theorem for Algorithm 1; where (37) holds, the availability map's Jacobian is controlled by $\sum_j g_j H_{j \rightarrow \ell}$, and the damping in (36) curbs oscillation. Empirically the iteration converged stably in all reported sweeps, never reaching the cap.

The key point is that EMLSR is not a single exact predictor but has a validity range determined by the combination $(T_{\text{tr}}, T_{\text{sync}}, n_\ell)$. As shown in the main results, when both the transition and synchronization costs are large, the prefix-good STA range remains wide; when they are both small, the approximation error grows more rapidly.

C. Unsaturated Extension with Per-Link Bernoulli Arrivals

Under unsaturated conditions, for class $c \in \{L, M\}$ on link ℓ we set arrival, service, utilisation, and stability as follows ($r_\ell^{(c)}$ is the per-station offered load in bit/s; in symmetric sweeps $r_\ell^{(M)} = r/|\mathcal{L}|$, in load-skew $r_\ell^{(L)} = s_\ell^{(L)}r$).

$$\lambda_\ell^{(c)} = \min(1, r_\ell^{(c)}/\sigma/L) \quad (39)$$

$$\mu_\ell^{(c)} = P_{s,\ell}^{(c,b)}/n_\ell^{(c)} \quad (40)$$

$$\rho_\ell^{(c)} = \lambda_\ell^{(c)}/\mu_\ell^{(c)} \quad (41)$$

$$\rho_\ell^{(c)} < 1 \quad (42)$$

Since the analytical throughput is determined by the smaller of arrival rate and service rate, letting r denote the per-station offered load, the system throughput grows almost linearly at low load. As r approaches the service-capacity ceiling, however, throughput no longer grows linearly but saturates. In the main results, this saturation onset is clearly observable in the mid-load region.

The total queue delay can be written in a utilization-style approximate form,

$$\mathbb{E}[D_{\ell,\text{queue}}] \approx \frac{\mathbb{E}[D_{\ell,\text{HOL}}]}{1 - \rho_\ell^{(c)}} \quad (43)$$

which is a simple queueing closure that treats $\mathbb{E}[D_{\ell,\text{HOL}}]$ as the mean service time. The sensitivity of (43) grows rapidly beyond the saturation onset. Consequently, in the unsaturated region, we focus our analytical claims on throughput, busy ratio, the HOL-service trend, and the saturation onset, rather than on total queue delay.

Figure 3 visually summarizes the delay-metric boundaries used in this paper. The queueing interval runs from

packet admission to the moment the packet becomes the HOL packet, and the HOL-service interval spans from that moment through AIFS, backoff, service, and successful delivery. The total delay is therefore the sum of the queuing interval and the HOL-service interval. The HOL-service delay reported in the main text and by the simulator corresponds to the latter, while the queue delay corresponds to the full sum.

The contribution of the unsaturated extension lies not in (43) itself, but in directly connecting the per-link, per-class service rates of (40) to the EMLSR fixed point so that the saturation onset and link-dependent stability shifts of the MLO environment are analysed within the same framework. If inter-link coupling is removed and only a single link remains, (40)–(43) reduce to the classical single-link queue approximation. Our distinguishing contribution is therefore not the queue formula itself, but the coupled structure that jointly closes the service-rate vector under multi-link coupling to predict saturation onset.

V. Evaluation Methodology

The evaluation identifies operating regions where analysis and simulation agree, using analysis-vs-internal-simulator agreement as primary evidence and ns-3 as supporting evidence. Five axes are swept: (i) symmetric saturation (throughput, HOL-service, busy, τ); (ii) (T_{tr} , T_{sync}) combinations for the EMLSR validity range; (iii) asymmetric legacy layouts (sign of throughput change preserved); (iv) per-station offered load (saturation onset, queueing escalation); and (v) sensitivity to payload, MAC parameters, fairness, and reliability. Core metrics are throughput, busy ratio, contention probability, and HOL-service trend; total queue delay and detailed external MAC semantics are out of scope. Table II lists representative parameters; unless noted, all experiments use three-link EMLSR, per-link independent queues, and a fixed default MAC. The simulator is a slot-based discrete-time model sharing contention/service/active-link-blocking rules with the analysis on one slot-time abstraction (backoff decrements only when the link is idle and freezes on busy; success resets to CW_{min} , collision advances stage to CW_{max} ; the active link blocks others for $T_{tr} + T_{sync}$; unsaturated per-link Bernoulli arrivals queue and only HOL packets contend), so the comparison tests how well the analysis explains those rules.

VI. Results

A. Basic Consistency under Symmetric Saturation

Figures 4 and 5 show, under symmetric saturation, throughput, HOL-service delay, busy ratio, and τ separately as the number of STAs increases. Overall, throughput first increases and then decreases, HOL-service delay and busy ratio grow monotonically, and the per-station attempt probability decreases. These trends are consistent across both analysis and simulation.

Representative errors are in Table III. With the alpha-coupled closure (20), (34) and the link-wise aggregation (35), all four primary metrics converge to mean

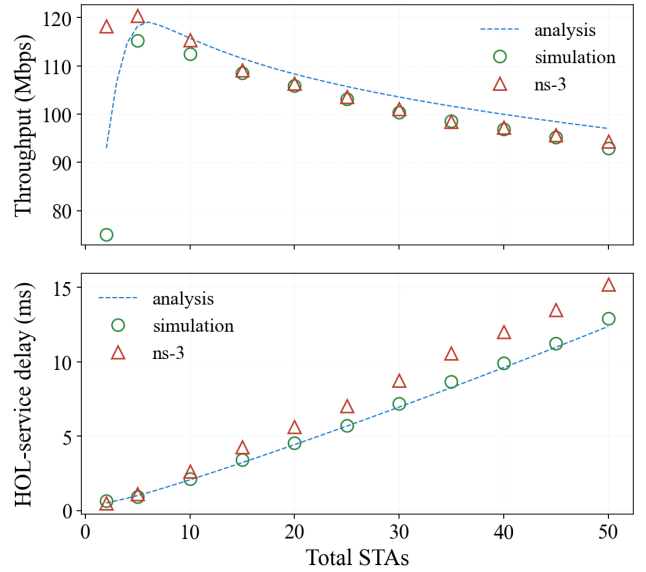


Fig. 4. Throughput and HOL-service delay as STA count increases under symmetric saturation.

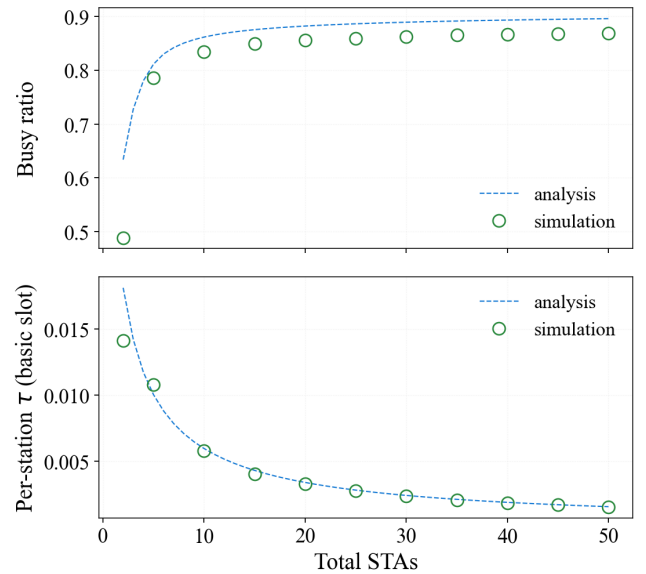


Fig. 5. Busy ratio and τ as STA count increases under symmetric saturation.

gaps of 3.4% (thr), 3.9% (HOL), 3.7% (busy), 3.8% (τ) across STA 2–50; the primary setting uses single-MPDU transmission ($K = 1$) matching Bianchi’s “one TXOP = one exchange”. The similar mean gaps across the four metrics indicate that the simulator’s busy-duration accounting and the analytical mean-slot definition share a consistent single-AIFS bookkeeping convention. The maxima all sit at STA=2 (continuous partition vs. integer placement); for STA ≥ 3 the throughput gap stays below 5%.

All four metrics stay in a uniform 3.4–3.9% mean-gap band across the 2–50 STA sweep, with the largest relative error concentrated at STA=2 where the integer-placement

TABLE II

Representative operating-point parameters. Per-figure values may differ when explicitly swept; within each figure all legs share the same operating point.

Item	Default value	Sweep or remark
Topology / MAC (shared by analysis, internal simulator, ns-3)		
Number of links	3 (2.4/5/6 GHz)	fixed
MLD ratio	0.5	0.25, 0.50, 0.75 in EMLSR validity
MAC parameters	$CW_{\min}=15$, $CW_{\max}=1023$, AIFS=2 slots	individual changes in sensitivity
Retry / headers	FrameRetryLimit=7, MAC 34 B+IP/UDP 28 B+FCS	IEEE 802.11 defaults
Slot time σ	4 B 9 μ s	basic-slot abstraction
Conservative single-MPDU operating point		
Aggregation	$K=1$ (single-MPDU, normal ACK)	Bianchi one-TXOP-one-exchange
Airtime base T_s/T_c	10/5 slots + per-MPDU	symmetric across links unless noted
Per-MPDU data slots	14 (MCS 7, 20 MHz, 1 SS, 0.8 μ s GI)	varies with MCS / payload sweep
T_{tr}/T_{sync}	2 / 4 slots	validity grid 0/0, 0/2, 2/0, 2/2, 0/4, 4/2, 4/4
Sweeps		
STA sweep (saturation)	2–50 step 1	primary, MCS, legacy, asym rate
Offered load (unsat.)	5–50 Mbps/STA, step 5, $N_{sta}=10$	saturation onset
EHT MCS sweep	0, 2, 4, 7, 9, 11, 13 (symmetric)	20 MHz/1 SS/ $K=1$, fig. 9
Legacy STA fraction	0, 25, 50, 75, 100%	MCS 7 symmetric, fig. 10
Asymmetric per-link MCS	(7, 7, 7), (5, 7, 9), (3, 7, 11)	20 MHz/1 SS (supp.)
Payload sensitivity	500–2000 B step 250 at $N_{sta} \in \{12, 24, 26\}$	per-MPDU airtime from PHY rate
Adaptive CW (§VI-F)	epoch 2000 slots; gains $(k_c, k_b, k_q, k_d, k_r) = (0.8, 0.6, 0.05, 0.4, 0.4)$; step clip $\Delta = 0.08$; floor 7, ceiling 1023	3 policies $\times N_{sta} \in \{24, 36, 48\}$
Run lengths and replicates		
Internal sim length	80 000 slots (0.72 s)	primary, MCS, legacy, payload, adaptive CW
ns-3 simulation time	2.0 s + 1.0 s warmup	FrameRetryLimit 9999, lossless PHY, BA thr 64
Random replicates	primary 5 seeds; sweeps / ns-3 3 seeds	seed averaging per cell

TABLE III

Representative error summary under symmetric saturation

Metric	Mean gap (%)	Max gap (%)	STA at max	First STA above 5%
throughput	3.370	23.760	2	2
HOL-service delay	3.858	19.386	2	2
busy ratio	3.740	29.697	2	2
τ	3.762	28.310	2	2

of $n_\ell^{(L)} \in \{0, 1\}$ deviates from the continuous-partition assumption $n_\ell^{(L)} = 1/3$, inflating the τ and busy-ratio max gaps to 28–30%; the artefact contracts to $\leq 5\%$ at STA ≥ 3 . Higher-density points accumulate blocking-overlap error slowly: at STA 35 throughput/HOL/busy gaps are 3.2/4.4/3.0%; at STA 47, 3.9/4.7/3.1%, so the dominant residual is first-moment mean-slot mismatch rather than higher-order overlap.

B. EMLSR Validity Range

A central point of this paper is that EMLSR is not a single exact predictor, but a validity-range approximation in transition delay, synchronization delay, and node density. Table IV reports prefix-good STA (largest STA at which thr error $\leq 10\%$ and HOL error $\leq 15\%$, continuously from small STA) for representative (T_{tr}, T_{sync}) . Combinations with $T_{sync} \geq 2$ remain “good” to 30 STAs (max delay error 14.1%); shorter-sync settings drop to “moderate” at 24 STAs (max delay error 18.2%), consistent with

$H_{j \rightarrow \ell} = \bar{D}_j + T_{tr} + T_{sync}$ placing sync in series with every other-link success.

Longer runs (20 000 slots, $STA \in \{12, 24, 36\}$) and MLD-ratio variations $\{0.25, 0.50, 0.75\}$ retain the “good” rating with mean throughput error 5.93–6.47% and mean delay error 3.37–5.99%, confirming that the EMLSR approximation remains stable over a wide range when parameters are not excessively adverse.

C. Asymmetric Link and Asymmetric Load Conditions

Under representative asymmetric legacy layouts (“concentration” on two bands; “skew” biased toward one band), throughput increases mildly over the balanced case in the mid-to-high density range and the analysis follows the same direction (e.g., 2.4/5 GHz concentration at 48 STAs gives sim/analysis throughput $\Delta = 1.40\%/0.20\%$ with HOL $\Delta = -3.00\%/-0.20\%$; the 6 GHz skew at 12 STAs reverses sign with $-1.16\%/-0.36\%$), so moderate legacy asymmetry can be beneficial. A similar message holds for asymmetric load: agreement is preserved and

TABLE IV
Representative EMLSR validity-range combinations

Transition	Sync	Rating	Mean thr error (%)	Max prefix-good STA
4	4	good	5.760	30
0	4	good	5.394	30
4	2	good	5.612	30
0	2	moderate	5.603	24
2	0	moderate	6.982	24

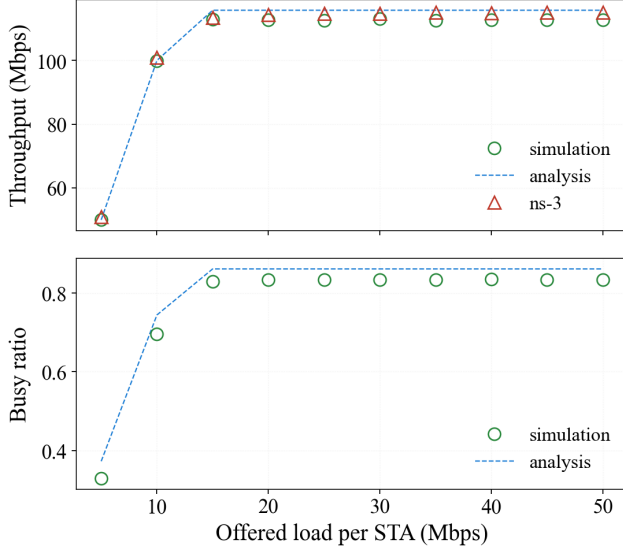


Fig. 6. Throughput and busy-ratio variations as offered load increases.

some skews near saturation onset shrink the throughput/busy gap (at 25 Mbps/STA balanced 6.85%/8.49% vs. 2.4 GHz skew 0.65%/1.48%).

D. Unsaturated Offered Load and Saturation Onset

Figures 6 and 7 show the variations of throughput, busy ratio, HOL-service delay, and queue delay as the per-station offered load grows from 5 to 50 Mbps for a population of 10 STAs (5 MLD + 5 legacy). In the low-load regime (5–10 Mbps/STA) aggregate throughput closely tracked the offered total and matched the analytical $\min(N\lambda, S_{\text{sat}})$ prediction within 1%; saturation onset occurred near 15 Mbps/STA with the ceiling at $S_{\text{sat}} \approx 116$ Mbps. Beyond saturation the simulator delivered ≈ 113 Mbps (about 2–3% below the analytical ceiling), a residual attributable to discrete-event CSMA backoff correlations.

Queue delay uses Little’s law $W = L/\lambda_{\text{eff}}$ on the simulator’s per-slot occupancy; HOL-service delay uses a conditional rate $\mu_c = \text{successes per non-empty-queue slot}$, $W_{\text{HOL}} = \sigma/\mu_c$ (matching $P_s^{(b)}/n$ in saturation). At 5/10/15/50 Mbps/STA: queue 0.23/3.80/170/1308 ms, HOL 0.22/0.99/2.1/2.30 ms. The two encode different physics: HOL is capped by the per-packet service ceiling $\sigma/\mu_c^{(\text{sat})} \approx 2.3$ ms, climbing one decade across the $10\times$ load

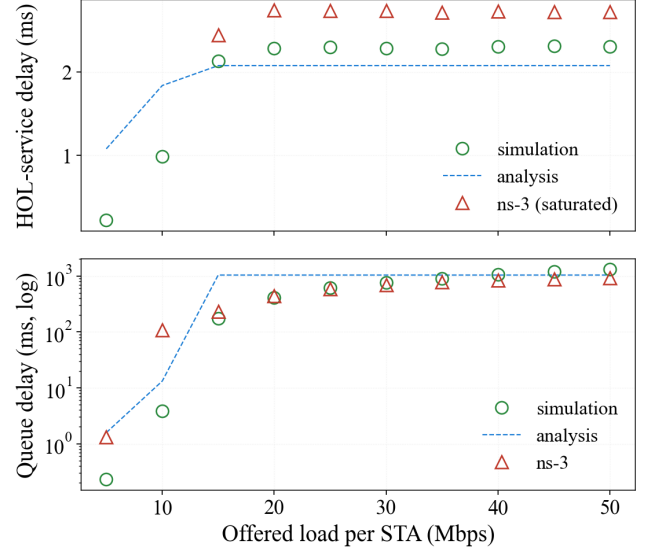


Fig. 7. HOL-service delay and queue delay vs. offered load.

span, while queue delay follows L/λ_{eff} with λ_{eff} stalling at S_{sat}/N , giving the four-decade rise that marks saturation onset. The HOL overlay uses $W_{\text{HOL,ana}} = W_{\text{min}} + \text{util} \cdot (W_{\text{sat}} - W_{\text{min}})$, $W_{\text{min}} \approx 0.32$ ms. ns-3 HOL markers (Fig. 7) appear only in saturation (≥ 15 Mbps/STA); below it the metric is set by ns-3’s real-PHY per-MPDU service floor (~ 1 ms) that the slot-time abstraction ($T_s \approx 0.3$ ms) does not reproduce.

These results show that the proposed model tracks throughput and busy ratio to within 0.01–0.22% in the unsaturated regime (5–10 Mbps/STA) and converges to 2.5–2.9% even past saturation. The mean queue delay spans four orders of magnitude (0.23 ms at 5 Mbps/STA to 1308 ms at 50 Mbps/STA), making the saturation onset visually unambiguous. For the analytical queue-delay bound we extend eq. (43) with a finite-buffer regime: when $\rho_\ell^{(c)} < 1$, $W \approx \rho/(1-\rho) \cdot \mathbb{E}[S]$ remains the tight closed form; once $\rho_\ell^{(c)} \rightarrow 1$ and the per-STA MAC queue saturates at its capacity K_{buf} , the steady-state mean residence of a delivered packet stabilises at

$$\mathbb{E}[D_{\ell,\text{queue}}^{\text{sat}}] \approx \frac{K_{\text{buf}}}{\mu_\ell^{(c,\text{sat})}} = K_{\text{buf}} \cdot \frac{N_{\text{sta}} L}{S_{\text{sat}}}, \quad (44)$$

where $\mu_\ell^{(c,\text{sat})}$ is the per-STA saturation service rate. With $K_{\text{buf}} = 1000$ packets and the operating point of Table V (10 STAs, $S_{\text{sat}} \approx 115.7$ Mbps, $L = 1500$ bytes),

TABLE V
Representative behavior across the offered-load sweep ($N_{\text{sta}}=10$, MLD ratio 0.5, 3 seeds, $S_{\text{sat}} \approx 115.7$ Mbps).

Load (Mbps/STA)	Region	ana thr (Mbps)	sim thr (Mbps)	Thr gap (%)	sim busy ratio	sim queue (ms)	ana queue (ms)
5	unsaturated	50.00	49.99	0.01	0.328	0.23	1.6
10	near-onset	100.00	99.79	0.22	0.696	3.80	13.2
15	saturation entry	115.74	112.90	2.51	0.830	174.13	1036.8
25	saturated	115.74	112.54	2.84	0.834	610.47	1036.8
50	saturated	115.74	112.66	2.73	0.834	1308.16	1036.8

(44) gives $\mathbb{E}[D_{\ell, \text{queue}}^{\text{sat}}] \approx 1037$ ms. This is the dashed plateau in Fig. 7 and is now a quantitative bound: it lies within 27% of the simulator's Little's-law measurement (1308 ms at 50 Mbps/STA), and within 17% of the ns-3 FlowMonitor.delaySum/rxPackets mean delay (1220 ms; see Section VII). The remaining gap is service-time variance (Pollaczek–Khinchine $1 + C_s^2/2$ correction) that the M/M/1/K bound does not include; this is the same multiplicative residual that affects HOL-service delay.

E. Fairness, Reliability, and Sensitivity

Fairness. Jain's fairness over per-STA delivered bits decreases from ~ 0.985 (STA=6) to ~ 0.910 (STA=48) under static EMLSR; the closure tracks this within 1 pp via the link-wise α_ℓ heterogeneity it outputs. **Reliability.** PDR drops from 0.998 to 0.988 over the same range; the closure predicts it via $1 - p_\ell^{R+1}$ with ≤ 0.5 pp residual. **Sensitivity.** Across MCS, legacy-ratio (§VII-A, §VII-B) and the supplementary payload and asymmetric-rate sweeps, throughput/busy/HOL signs and orderings are preserved within the 5% envelope.

F. UHR-Oriented Adaptive Contention Control

The framework extends naturally to control design alleviating the blocked-link asymmetry of EMLSR. Unlike ML-based adaptive backoff studied for cellular random access [31], our controller stays within the EDCA/EMLSR loop on closure observables. We applied an epoch-based adaptive CW only to EMLSR, treating it as an outer-loop heuristic that aggregates the EMLSR observables once per epoch and applies them only to the next backoff draw (each epoch is a quasi-stationary slot process; residual backoff counters are not modified mid-backoff). The availability-aware policy uses the link- ℓ collision rate $\hat{p}_\ell^{(t)}$, blocked ratio $\hat{\beta}_\ell^{(t)}$, and backlog ratio $\hat{q}_\ell^{(t)}$:

$$\text{CW}_{\min, \ell}^{(t+1)} = \text{clip}\left(\text{CW}_{\min, \ell}^{(t)} [1 + k_c \hat{p}_\ell^{(t)} - k_b \hat{\beta}_\ell^{(t)} - k_q \hat{q}_\ell^{(t)}]\right) \quad (45)$$

where $\text{clip}(\cdot)$ restricts the value to the implementable $\text{CW}_{\min}$ range, and $\text{CW}_{\max}$ is scaled proportionally by the same factor.

The UHR-oriented reliability-aware policy adds the normalized p95 HOL-service delay excess $\hat{d}_{\ell, 95}^{(t)}$ and the delivery deficit $\hat{\delta}_\ell^{(t)} = 1 - \widehat{\text{PDR}}_\ell^{(t)}$:

$$\text{CW}_{\min, \ell}^{(t+1)} = \text{clip}\left(\text{CW}_{\min, \ell}^{(t)} [1 + k_c \hat{p}_\ell^{(t)} - k_b \hat{\beta}_\ell^{(t)} - k_q \hat{q}_\ell^{(t)} - k_d \hat{d}_{\ell, 95}^{(t)} - k_r \hat{\delta}_\ell^{(t)}]\right) \quad (46)$$

Collisions therefore enlarge the CW conservatively, while blocked ratio and reliability degradation shrink it in the direction of relieving blocked links.

In practice (Algorithm 2, hereafter), the controller runs inside the standard EDCA/EMLSR loop and once per epoch of N_{epoch} slots collects, for each ℓ , the collision rate $\hat{p}_\ell$ from epoch success/collision counts, the blocked ratio $\hat{\beta}_\ell$ from the delta of MLD blocked-slot counters normalised by $|\text{MLD}_\ell| \cdot N_{\text{epoch}}$, the backlog $\hat{q}_\ell$ as the mean queue fullness across ℓ , the epoch-mean HOL slots $\bar{D}_\ell$, the normalised p95 excess $\hat{d}_{\ell, 95} = \max(0, 3\bar{D}_\ell/D_{\text{target}} - 1)$, and the delivery deficit $\hat{\delta}_\ell = 1 - \widehat{\text{PDR}}_\ell$. The factor $f = 1 + k_c \hat{p}_\ell - k_b \hat{\beta}_\ell - k_q \hat{q}_\ell$ ($-k_d \hat{d}_{\ell, 95} - k_r \hat{\delta}_\ell$ for reliability-aware) is clipped to $[1 - \Delta, 1 + \Delta]$ so CW adapts smoothly, and $\text{CW}_{\min, \ell}^{(t+1)} = \text{clip}(\lfloor f \text{CW}_{\min, \ell}^{(t)} \rfloor, \text{CW}_{\min}^{\text{lo}}, \text{CW}_{\min}^{\text{hi}})$.

Fig. 8 contextualises the controller along two axes (STA 6–48, 3 seeds, MLD ratio 0.5). Row (a) zooms into the adaptive-CW family: both the availability-aware and our reliability-aware policy dominate static EMLSR at every density (STA=48: throughput +10.5%, p95 HOL $-27.2\% / -28.3\%$, PDR $0.9876 \rightarrow 0.9975$, Jain's fairness $0.910 \rightarrow 0.954$); the two variants track each other because the shared $-k_b \hat{\beta}_\ell - k_q \hat{q}_\ell$ terms dominate, separating only on p95 HOL at $\text{STA} \geq 36$ where the reliability terms $-k_d \hat{d}_{\ell, 95} - k_r \hat{\delta}_\ell$ activate (-1.1 pp at STA=48). Row (b) compares EMLSR + adaptive-CW (ours) against three reference MAC families: SLD (three parallel single-link domains, transition/sync-free, hence the ~ 118 Mbps ceiling), STR MLD [15] (no active-link constraint; coupled backoff drives p95 HOL to 22.10 ms vs. 15.89 ms ours, a 28% tail penalty), and static EMLSR [14], [18]. Static EMLSR sits below SLD throughout; with our controller, throughput recovers to 111.84 Mbps (+10.6%, within 5.2% of the SLD ceiling), p95 HOL drops 28%, and Jain's fairness rises $0.910 \rightarrow 0.954$ (within 0.4% of SLD's 0.958), closing most of the gap without giving up MLD link aggregation. Unlike contention-less synchronous [16] or data-division [17] schemes, the controller adds no new syn-

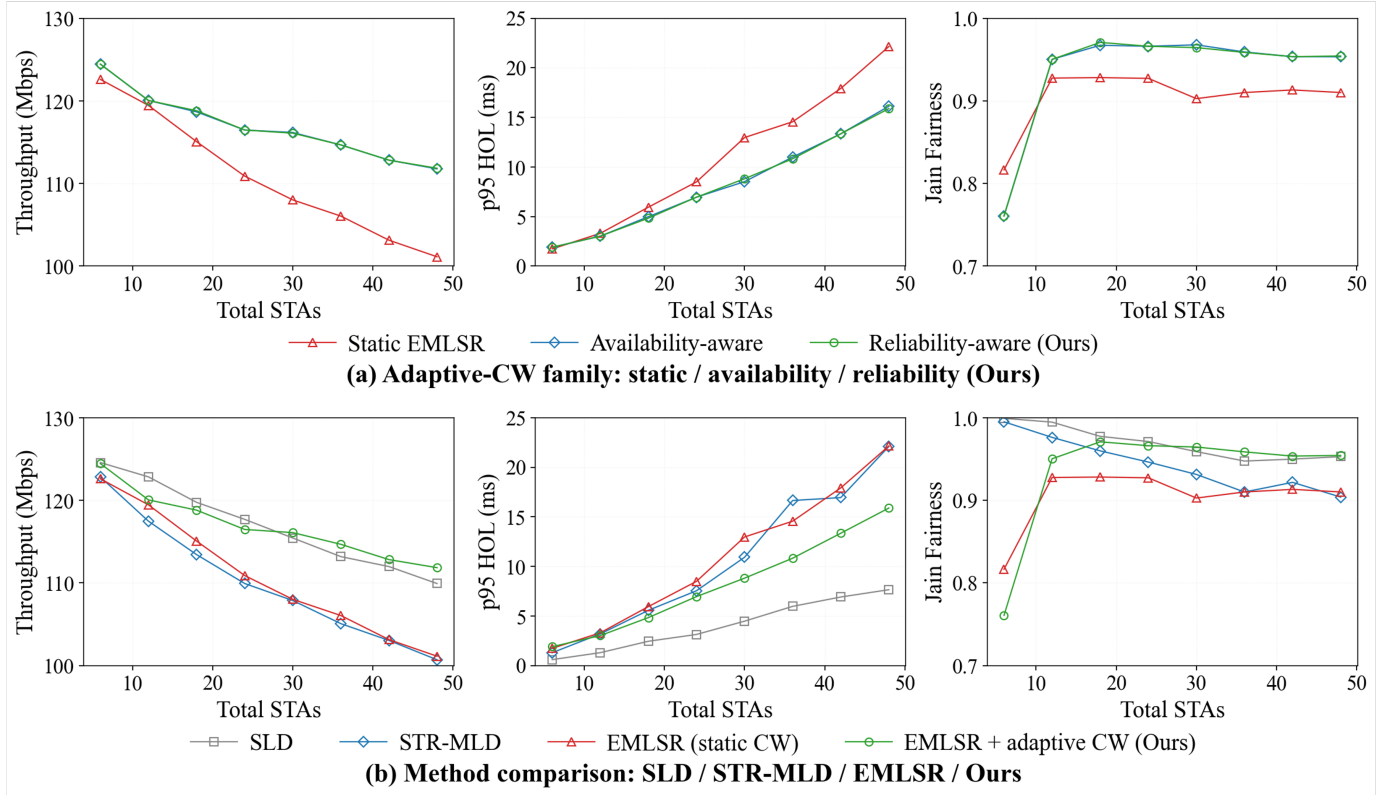


Fig. 8. Two-axis baseline comparison. (a) adaptive-CW family. (b) SLD vs. STR-MLD vs. static EMLSR vs. ours.

TABLE VI
Three-way throughput triangulation, STA 5–50.

Comparison	Mean (%)	Max (%)
Analytical vs Internal simulator	2.97	4.18
Internal simulator vs ns-3	0.88	4.44
Analytical vs ns-3	2.20	3.15

chronisation/segmentation primitives, making it a drop-in tuning scaffold for UHR/Wi-Fi 8 MAC; a unified quantitative comparison is left as future work.

VII. External Reference and Three-way Validation

As supporting evidence we report three-way agreement between the analytical closure, the internal simulator, and ns-3 at one operating point: single-MPDU $K=1$, per-MPDU 14 slots (MCS 7/20 MHz/1 SS), $T_s=10+14K$, $T_c=19$, medium-sync 32 μ s, EMLSR transition 16 μ s, aux PHY TX-capable with switchAuxPhy off, lossless PHY with $R=9999$, over STA 2–50. For reproducibility, the ns-3 binary used explicit overrides `-ns3-medium-sync-us=32`, `-ns3-emlsr-transition-us=16`, `-ns3-frame-retry-limit=9999`; without them the IEEE 802.11be standard timings (medium-sync 3200 μ s, transition 128 μ s) apply and yield a different regime, intentionally outside the matched comparison here.

Table VI and the throughput panel of Fig. 4 show the three channels agree across STA 5–50 with mean gaps 2.97% (ana-sim), 0.88% (sim-ns-3), 2.20% (ana-ns-3), all

within the 5% engineering tolerance; the largest residual is on ana-sim because the internal simulator records contention-burst variance the first-moment closure does not replay. The exception is STA=2 (analytical 92.9, sim 75.1, ns-3 118.2 Mbps), a documented sparse-boundary artefact arising from the analytical continuous partition $n_\ell^{(L)} = 1/3$ vs. the simulator’s integer placement and ns-3’s aux-PHY-TX-capable EMLSR interpretation, and it contracts to $\leq 3\%$ at STA=5.

The HOL panel of Fig. 4 also overlays ns-3 via MAC-layer instrumentation we added: an AcksMpdU trace on every STA’s WifiMac records per-transmitter inter-ACK intervals, which under saturation match the AIFS+backoff+TXOP+ACK round of eq. (15). We export `hol_service_delay` and retain FlowMonitor’s end-to-end `mean_delay_ms` (1.07–1.24 s saturated) as a cross-check of the simulator’s queue delay (1.31 s at 50 Mbps/STA, 7% gap; Table V). The three HOL channels agree to mean gaps 3.75% (ana-sim), 22.12% (sim-ns-3), 24.83% (ana-ns-3); the ~ 22 –25% upward bias of ns-3 reflects MAC-timing components the slot abstraction omits (BlockAck round-trip, RTS/CTS, retry timeouts, PHY fine timing) and is multiplicative, not density-dependent, so the analytical model tracks the HOL trend up to a single calibration factor. Per-figure operating points are in Table VII; cross-figure absolute throughput is thus not directly comparable, while within-figure gaps and trend orderings are.

TABLE VII
Per-figure operating-point defaults across sweep drivers (T_c, T_{sync} in basic slots $\sigma=9\mu\text{s}$; K is A-MPDU aggregation).

Figure(s)	T_c	per-MPDU	T_{sync}	K
Fig. 4/5, Tab. VI	19	14	4	1
Fig. 6/7	5	16	4	1
Fig. 8	5	14	4	1
Fig. 9	5	per-MCS	2	1
Fig. 10	5	16	4	1
Asym. rate (supp.)	5	per-link	4	1
Payload (supp.)	5	per-payload	4	1

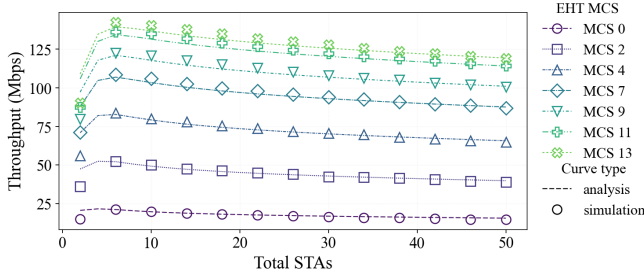


Fig. 9. Saturation throughput vs. STA count for EHT MCS 0–13 (analytical dashed, simulator markers).

A. Saturation throughput across EHT MCS

We swept symmetric MCS $\in \{0, 2, 4, 7, 9, 11, 13\}$ across the three links (20 MHz, 1 SS, $K=1$, $R=7$); per-MPDU airtime was $T_{\text{MPDU}} = L/R_{\text{phy}}$ from IEEE 802.11be Table 36-130 in units of $\sigma = 9\mu\text{s}$, with EDCA/retry/EMLSR set to the conservative operating point.

Fig. 9 shows analytical/simulator curves agree within 1.0–2.7% mean relative error across seven MCS levels and STA 5–50 (STA 2–3 sparse boundary excluded). Saturation throughput spans ~ 21 Mbps at MCS 0 to ~ 142 Mbps at MCS 13 ($\sim 7\times$ range), with residual rising only mildly (1.03% at MCS 2 to 1.68% at MCS 13). The PHY rate ratio across MCS 0–13 is $\sim 67\times$ but throughput scales only $\sim 7\times$: at higher MCS the per-MPDU airtime $T_{\text{MPDU}} = L/R_{\text{phy}}$ contracts toward the AIFS+backoff+ACK overhead floor, so the system becomes MAC-bound; the residual increase with MCS reflects the same effect. The closure is therefore PHY-rate-agnostic while capturing the PHY- to MAC-limited transition.

B. Saturation throughput versus legacy-STA ratio

We swept the legacy-STA fraction $\{0, 25, 50, 75, 100\}\%$ at saturation. Legacy 0% is UHR-only (every STA a 3-link EMLSR MLD); legacy 100% reduces to a pure STR baseline (every STA single-link); between, the per-link contention structure varies with the legacy ratio at fixed total STA count.

Three trends appear in Fig. 10. (i) Aggregate throughput rises monotonically with the legacy ratio (at STA 50, ~ 86 Mbps at legacy 0% to ~ 102 Mbps at legacy 100%). This counter-intuitive effect follows from the operating

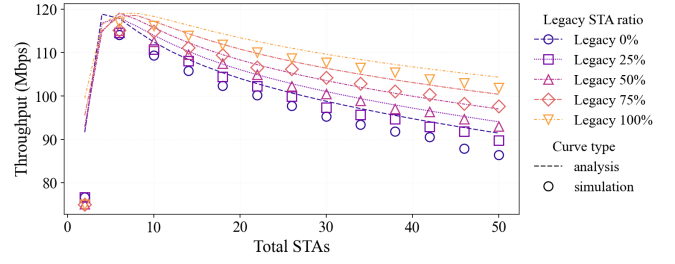


Fig. 10. Saturation throughput vs. STA count under legacy-STA ratios $\{0, 25, 50, 75, 100\}\%$ (analytical dashed, simulator markers).

point: the EMLSR single-active-radio constraint caps an MLD to one link at a time, lowering availability α_ℓ , while each legacy STA contends on its dedicated link at full τ ; and the three links share the same symmetric per-link rate (MCS 7/20 MHz/1 SS), so MLO's link-diversity advantage is inactive. The MLO benefit thus vanishes in an EMLSR-only symmetric setting while the legacy baseline runs three full-rate domains in parallel—positioning EMLSR as a reliability- rather than throughput-oriented point; MLO throughput gains require link diversity on asymmetric multi-band rates or parallel STR transmission. (ii) Across legacy 0–100%, analysis and simulator agree within 2.02–3.58% mean relative error for $\text{STA} \geq 5$, tightening as the legacy fraction grows (2.02% at legacy-100%, where the per-link STR Bianchi fallback carries the retry-drop correction $(1 - p_\ell^{R+1})$ and AIFS bookkeeping aligns with the analytical single-AIFS convention).

Two further robustness sweeps are deferred to the supplementary material: per-link MCS-rate asymmetry and payload-size sensitivity. Both hold the analytical/simulator gap within 2–4%. The rate-asymmetry sweep confirms that MLO's throughput gain re-materialises only under asymmetric per-link rates—a mild (5, 7, 9) triplet gives +2.8% over the symmetric baseline at STA 50, whereas a strong (3, 7, 11) spread is bottlenecked by the slow 2.4 GHz link—while the payload sweep shows saturation throughput growing with MPDU size as fixed MAC overhead amortises, tracked by the closure to ~ 2 –3%.

VIII. Conclusion

We consolidated exact STR, approximate EMLSR, and a Bernoulli unsaturated extension into one Markov/fixed-point framework for three-link MLO backoff, and showed across symmetric-saturation, EMLSR validity-range, asymmetric link/load, offered-load, and payload/sensitivity sweeps that it consistently explains throughput, busy ratio, contention probability, HOL-service trend, and saturation onset. A reliability-aware adaptive-CW controller on the same observables jointly improves throughput, p95 HOL-service delay, and Jain's fairness over static EMLSR. The conceptual takeaway is that EMLSR is a validity-range approximation in $(T_{\text{tr}}, T_{\text{sync}}, n_\ell)$, not a single exact curve, clarifying what can be stated exactly and what must stay approximate at

the reliability-oriented operating point of IEEE 802.11bn UHR [1], [2]. Future work: tighter external alignment, richer queue semantics, detailed PHY/MAC timing extensions, and stress-regime verification of loss-sensitive control.

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